

Designing Evidence-Traceability Metrics for Agentic Literature Synthesis Pipelines

Abstract. Agentic pipelines that autonomously search, retrieve, filter, and synthesize scholarly literature are increasingly proposed as accelerants for evidence-based research, yet their adoption is constrained by a well-documented tendency of large language models (LLMs) to fabricate citations, misattribute findings, and generate claims that are only loosely coupled to retrieved evidence. This paper examines the problem of designing *evidence-traceability metrics*—quantitative measures of the degree to which each claim produced by an agentic literature-synthesis system can be verifiably linked to a specific, existing, and semantically supportive source passage. We synthesize work from three previously disconnected communities: retrieval-augmented generation (RAG) attribution evaluation (e.g., ALCE, RAGAS, ARES, FactScore), citation-hallucination auditing in scientific and medical contexts, and evidence-synthesis automation in systematic-review methodology (e.g., RobotReviewer, ASReview). We propose a layered taxonomy of traceability metrics spanning *existence verification*, *attribution precision/recall*, *claim-evidence entailment*, and *provenance completeness*, and we discuss how multi-agent orchestration patterns intersect with each layer. We then review current tool-level and benchmark-level approaches, identify methodological gaps—including the absence of standardized, domain-general traceability benchmarks for multi-hop and multi-document synthesis, the confound between fluency and faithfulness, and the scalability limits of natural-language-inference (NLI)-based judges—and outline directions for future research, including reinforcement-learning-based traceability optimization, cross-agent provenance graphs, and standardized reporting norms modeled on PRISMA-style checklists. The paper is intended as a conceptual and methodological reference for researchers designing evaluation protocols for agentic research-synthesis systems.

Keywords: evidence traceability, agentic AI, retrieval-augmented generation, citation hallucination, literature synthesis, attribution evaluation, systematic review automation, multi-agent systems

1 Introduction

The volume of scholarly output continues to outpace the capacity of individual researchers and review teams to read, appraise, and synthesize it. Estimates from the evidence-based medicine literature have long noted that conducting a single systematic review can require months of specialist effort, motivating a decades-long research program in (semi-)automated evidence synthesis (Marshall, Kuiper, & Wallace, 2016; Khalil, Ameen, & Zarnegar, 2022). The recent emergence of large language models (LLMs) capable of reading, summarizing, and generating fluent prose over scientific text has intensified interest in agentic literature-synthesis pipelines: systems in which one or more LLM-driven agents autonomously plan search queries, retrieve candidate documents, filter and rank them, extract findings, and compose a synthesized narrative, often with iterative self-critique or multi-agent division of labor (Sumers, Yao, Narasimhan, & Griffiths, 2024; Gottweis et al., 2025).

This automation promise is shadowed by a specific and well-replicated failure mode: LLMs generate citations and factual claims that are fluent, plausible, and frequently wrong. Studies auditing LLM-generated references have found hallucination rates exceeding 25% even for models given explicit literature-search tasks (Chelli et al., 2024), and large-scale audits of LLM-generated citations across disciplines report substantial and, in some analyses, worsening rates of non-existent references (Sakai, Kamigaito, & Watanabe, 2026 [as cited in downstream audits]). Because a literature synthesis is only useful insofar as its claims can be checked against the sources that ostensibly support them, the central engineering and evaluation problem for agentic literature-synthesis pipelines is not merely factual accuracy in the abstract, but *traceability*: the ability to map every asserted claim, quantitatively and verifiably, back to the passage(s) of retrieved literature that license it.

This paper addresses the question of how to design metrics that measure evidence traceability in agentic literature-synthesis pipelines. We make three contributions. First, we synthesize related work across three literatures that have developed largely independently—RAG attribution evaluation, citation-hallucination auditing, and systematic-review automation—to establish a shared vocabulary. Second, we propose a layered taxonomy of traceability metrics (existence, attribution quality, claim-evidence entailment, and provenance completeness) and map current tools and benchmarks onto this taxonomy. Third, we identify concrete methodological gaps and propose directions for future work, including standardized benchmarking practices adapted from evidence-based medicine reporting norms.

The remainder of the paper is organized as follows. Section 2 reviews background and related work. Section 3 presents the proposed taxonomy and technical discussion of traceability metrics. Section 4 surveys current approaches and systems. Section 5 discusses research challenges and limitations. Section 6 identifies research gaps and future directions. Section 7 concludes.

2 Background and Related Work

2.1 Retrieval-Augmented Generation and the Attribution Problem

Retrieval-augmented generation (RAG) was introduced as a general recipe for combining a parametric language model with a non-parametric retrieval index, with the explicit motivation that non-parametric memory could provide *provenance* for generated content and allow world knowledge to be updated without retraining (Lewis et al., 2020). This provenance motivation is central to the traceability problem: RAG was proposed partly *because* standalone parametric LLMs cannot indicate which of their outputs are grounded in verifiable sources.

Subsequent work formalized what it means for generated text to be attributable to a source. Rashkin et al. (2023) introduced the Attributable to Identified Sources (AIS) framework, which evaluates whether a generated statement s would pass the test “according to source P , s ”—that is, whether a generic reader, given access to P , would agree that P entails s . AIS deliberately separates attribution from other quality dimensions such as fluency or informativeness, and has become a conceptual reference point for later attribution benchmarks (Bohnet et al., 2022, as adapted for QA; Gao, Yen, Yu, & Chen, 2023).

Gao et al. (2023) operationalized attribution evaluation for citation-augmented long-form generation with ALCE (Automatic LLMs’ Citation Evaluation), introducing *citation recall*—whether the concatenation of a sentence’s cited passages entails that sentence, typically assessed with a natural-language-inference (NLI) model such as TRUE (Honovich et al., 2022)—and *citation precision*—whether each individual citation is load-bearing, tested by removing it and checking whether entailment is lost. ALCE additionally reports fluency and answer correctness, and the framework spans three datasets (ASQA, QAMPARI, ELI5) that stress different citation regimes (short-answer, list-answer, and long-form explanatory generation, respectively). Follow-on frameworks have extended or complicated this picture: several works propose finer-grained or reference-free variants that probe automated NLI-style judges directly and add dimensions such as contradiction detection alongside support detection; RAGAS (Es, James, Espinosa-Anke, & Schockaert, 2024) and ARES (Saad-Falcon, Khattab, Potts, & Zaharia, 2024) shifted the evaluation target from citation-augmented QA specifically to general RAG-pipeline diagnostics, decomposing answers into atomic statements and scoring *faithfulness* (statement-to-context entailment), *context relevance*, and *answer relevance* without requiring gold references. ARES additionally uses prediction-powered inference to calibrate lightweight LLM judges against a small amount of human-labeled data, reporting substantial accuracy gains over RAGAS on context-relevance judgments (Saad-Falcon et al., 2024).

A parallel line of work developed fact-level rather than citation-level factuality metrics. FactScore (Min et al., 2023) decomposes long-form generations into “atomic facts” and scores the fraction supported by a

trusted knowledge source, showing that sentence-level scoring is too coarse: in their analysis, a substantial fraction of individual ChatGPT-generated sentences mixed supported and unsupported content, an important caution for any traceability metric that operates at the sentence rather than the proposition level.

2.2 Citation Hallucination in Scientific and Medical Contexts

A distinct but converging literature has directly measured citation hallucination in scholarly and biomedical writing tasks. Athaburi et al. (2023) manually validated 178 ChatGPT-generated references for medical research proposals against Google, Scopus, and PubMed, finding a minority were verifiably correct. Chelli et al. (2024), evaluating ChatGPT (GPT-3.5, GPT-4) and Bard against 11 human-conducted systematic reviews of shoulder rotator-cuff pathology (471 references total), reported precision rates as low as 0%–13.4% and an overall hallucination rate exceeding 25%, using a strict criterion in which a reference was deemed hallucinated if two or more of title, first author, year were incorrect (Chelli et al., 2024). Agrawal, Mackey, and Kalai (2024) probed hallucinated-reference detection across 200 computer-science topics, evaluating whether models could recognize their own fabricated titles when queried directly versus indirectly. Al-jumain et al. (2024) proposed a weighted Reference Hallucination Score (RHS) that assigns differential penalties to errors in different citation fields (title, authorship, venue, date) rather than treating hallucination as binary. Large-scale, retrieval-independent audits have since scaled this style of analysis: Sakai et al. (2026) manually and semi-automatically audited references across major NLP venues (ACL, NAACL, EMNLP) and reported a more than tenfold increase in the proportion of papers containing hallucinated citations between 2024 and 2025, evidence that citation hallucination is not confined to LLM outputs but is already propagating into the peer-reviewed literature itself. A separate large-scale audit of 111 million references across 2.5 million papers on arXiv, bioRxiv, SSRN, and PubMed Central found a conservative estimate of 146,932 hallucinated citations in 2025 alone, disproportionately concentrated in fields with rapid AI uptake and in manuscripts bearing linguistic signatures of AI-assisted writing, and reported that such hallucinated entries were beginning to appear embedded in downstream bibliographic infrastructure such as Google Scholar—raising the concern of a feedback loop in which future models are trained on today’s fabricated citations (Zhao, Wang, Stuart, De Vaan, Ginsborg, & Yin, 2026).

2.3 Systematic-Review Automation and Evidence-Based Methodology

Independently of the NLP attribution literature, the evidence-based medicine community has a mature tradition of (semi-)automating components of systematic review production, motivated by the same traceability concern but arrived at from a methodological-rigor rather than a machine-learning-evaluation direction. RobotReviewer (Marshall et al., 2016) uses supervised machine learning trained on the Cochrane Database of Systematic Reviews to extract PICO (population, intervention, comparator, outcome) elements and assess risk-of-bias domains directly from full-text randomized controlled trial (RCT) reports, always surfacing the supporting sentence(s) so a human reviewer can verify the system’s judgment against source text—an explicit, if narrow, traceability guarantee built into the tool’s interface. ASReview (van de Schoot et al., 2021) applies active learning to title/abstract screening, prioritizing likely-relevant records for human review rather than fully automating inclusion decisions, again preserving a human-verifiable link between system recommendation and source record. Khalil, Ameen, and Zarnegar (2022) scope 19 tools spanning screening and extraction phases and note that fully automated (rather than human-in-the-loop) systems remain the exception, largely because unverifiable automation in this domain carries direct clinical-guideline risk. This literature offers agentic-AI researchers a longer-standing normative baseline: in evidence-based medicine, an evidence-synthesis claim without a checkable source pointer is not merely lower-quality output but methodologically inadmissible.

2.4 Agentic and Multi-Agent Literature-Synthesis Systems

A newer wave of systems applies multi-agent LLM orchestration directly to literature review and synthesis. LitLLM (Agarwal, Wang, Rajwal, Kumar, & Nair, 2024/2025, published as “LitLLMs, LLMs for Literature

yet?”) reports that plan-based prompting combined with retrieval-augmented generation substantially reduces, but does not eliminate, citation hallucination in generated literature reviews. LiRA (Go, Ly, Søgaard, Tahatabaei, de Rijke, & Chen, 2025) is a multi-agent framework in which agents cite sources by full article title—used as a semantic anchor to reduce drift—before post-processing citations into numbered references, with hallucinated titles explicitly redacted at evaluation time. RAGentA (Besrou, He, Schreieder, & Farber, 2025) proposes a multi-agent RAG architecture for attributed question answering with iterative document filtering and completeness verification, reporting gains in both correctness and RAGAS-style faithfulness over single-pass RAG baselines. An open-source agentic hybrid-RAG framework for scientific literature review similarly argues that LLMs operating without reliable retrieval are “perilous” for review tasks and that integration with verifiable retrieval is a precondition, rather than an enhancement, for trustworthy synthesis. At the frontier of agentic science more broadly, systems such as Google’s AI co-scientist (Gottweis et al., 2025) and related multi-agent scientific-discovery frameworks apply generate-critique-rank-evolve loops to hypothesis generation grounded in literature review, illustrating that traceability requirements extend beyond summarization into claim generation for novel research proposals. Complementary reinforcement-learning-based approaches have begun to optimize traceability directly as a training signal rather than only a post-hoc evaluation criterion, rewarding models for producing outputs whose citations survive automated verification.

3 A Layered Taxonomy of Evidence-Traceability Metrics

Drawing on the literatures reviewed above, we propose organizing evidence-traceability metrics for agentic literature-synthesis pipelines into four layers, ordered from the weakest to the strongest traceability guarantee they certify. A pipeline can satisfy a lower layer while failing a higher one; we argue that reporting a single aggregate “accuracy” score, as is still common practice, obscures which layer is actually failing and therefore what engineering intervention is warranted.

3.1 Layer 1: Existence Verification

The minimal traceability requirement is that a cited source *exists*—that is, that the reference corresponds to a real, retrievable document with metadata (title, authors, venue, year, and a persistent identifier such as a DOI) that matches an authoritative index (e.g., Crossref, PubMed, Semantic Scholar, OpenAlex). Existence-verification metrics are typically reported as a hallucination rate: the proportion of generated citations that cannot be matched to a real document, or whose metadata mismatches an authoritative record on a threshold number of fields (Chelli et al., 2024 use a two-of-three field-mismatch criterion for title, author, and year). Existence verification is necessary but far from sufficient: a citation can point to a real paper that does not actually support the claim attached to it (a form of *topical* rather than *existence* hallucination), which existence checks alone cannot detect.

3.2 Layer 2: Attribution Precision and Recall

The second layer asks not merely whether a citation exists but whether the *cited passage supports the specific claim* attached to it, and whether *all* necessary support has been cited. This is the layer formalized by ALCE’s citation precision and recall (Gao et al., 2023) and generalized by RAGAS’s faithfulness metric (Es et al., 2024): given a decomposed claim (sentence or atomic-fact level) and its associated citation set, an NLI-style judge determines (i) recall—whether the union of cited passages entails the claim—and (ii) precision—whether every individual citation in the set is independently necessary, typically tested via leave-one-out entailment checks. Aggregating precision and recall (e.g., via an F1-style score) at the response level provides a single per-generation attribution score, but we note that reporting only the aggregate discards information about *where* in a synthesis the attribution failures concentrate, which matters for diagnosing whether a pipeline’s retrieval, ranking, or generation stage is the bottleneck.

3.3 Layer 3: Claim–Evidence Entailment and Semantic Fidelity

The third layer moves beyond binary citation–passage entailment to ask whether the *semantic contour* of the claim—not merely its literal support by a passage in isolation—is faithfully represented, including direction, magnitude, population, and conditionality of the underlying finding. This layer addresses a specific failure documented by evidence-based-medicine researchers: a claim can be technically entailed by a cited passage under a loose NLI judgment while still misrepresenting the finding’s scope (e.g., converting a subgroup result into an unqualified population-level claim, or omitting an effect-size confidence interval that changes the interpretation). Metrics here draw on atomic-fact decomposition (FAcTScore; Min et al., 2023) combined with stricter, structured claim extraction (e.g., converting findings to PICO-style tuples, as RobotReviewer does for source extraction) so that entailment can be checked at the level of qualified propositions rather than surface sentences. Recent work distinguishing *relevance* from *warrant* in cited RAG generation similarly argues that a passage’s topical relevance to a claim is a substantially weaker condition than its ability to license the specific inferential strength of that claim, and proposes force-calibration diagnostics as a complementary check to standard entailment scoring.

3.4 Layer 4: Provenance Completeness and Cross-Agent Traceability

The fourth and most demanding layer is specific to *agentic*, as opposed to single-pass RAG, pipelines: because an agentic literature-synthesis system typically involves multiple stages (query planning, retrieval, filtering/ranking, extraction, drafting, and often multi-agent critique or revision), a claim’s ultimate citation may be several transformation steps removed from the original retrieved passage. Provenance-completeness metrics ask whether the full chain—from original source passage, through intermediate extraction or summarization steps performed by upstream agents, to the final cited claim—can be reconstructed and audited, and whether errors introduced at any intermediate step (e.g., a summarization agent silently dropping a qualifying condition) are detectable. This layer is closely related to problems studied under the heading of agent hallucination more broadly, where hallucinations are understood as compound, multi-step, and potentially self-reinforcing failures that propagate and accumulate across an agent’s operating cycle rather than arising as single, localized generation errors. We are not aware of a widely adopted, general-purpose benchmark that specifically operationalizes provenance completeness for multi-agent literature synthesis (as opposed to single-turn RAG), which we discuss further as a research gap in Section 6.

Table 1: Layered taxonomy of evidence-traceability metrics, representative operationalizations, and typical failure modes detected.

Layer	Representative metric(s)	Failure mode detected
1. Existence	Reference hallucination rate; field-match against Cross-ref/PubMed/OpenAlex	Fabricated or non-resolvable citations
2. Attribution precision/recall	ALCE citation P/R; RAGAS faithfulness; ARES answer faithfulness	Claims cited to irrelevant or insufficient passages
3. Claim–evidence entailment	FAcTScore atomic-fact support; structured (PICO-style) entailment; force/warrant calibration	Overgeneralization, scope drift, magnitude distortion
4. Provenance completeness	Cross-agent provenance graphs (emerging; no standard benchmark)	Silent information loss across multi-agent transformation steps

4 Current Approaches and Developments

4.1 Reference-Based and Reference-Free Evaluation Frameworks

Two broad evaluation strategies coexist in current practice. *Reference-based* approaches, exemplified by ALCE, compare system output against gold citation sets or gold answers curated for a fixed benchmark (ASQA, QAMPARI, ELIS), enabling precise, reproducible scoring but limiting evaluation to the topical coverage of the benchmark. *Reference-free* approaches, exemplified by RAGAS and ARES, instead use LLM-based or fine-tuned NLI judges to score faithfulness against whatever context the system itself retrieved, at the cost of injecting judge-model bias and error into the very tool used to detect pipeline error. ARES’s use of prediction-powered inference—calibrating a small set of human labels against a much larger set of LLM-judged examples to obtain statistically corrected estimates—has been reported to outperform RAGAS’s uncalibrated approach by a substantial margin on context-relevance judgments across benchmark datasets spanning KILT and SuperGLUE-derived tasks (Saad-Falcon et al., 2024). Domain-specific evaluation resources have also emerged: ExpertQA specifically stress-tests attribution quality in high-stakes technical domains (medicine, law) using expert-authored questions rather than crowd-sourced ones.

4.2 Citation-Grounding Interventions at the Generation Stage

Beyond post-hoc evaluation, several current systems intervene directly at generation time to improve traceability. LIRA’s use of full article titles as in-generation semantic anchors, rather than opaque numeric placeholders, is intended to keep the generating agent’s attention causally coupled to the correct source throughout decoding, with hallucinated titles identified and redacted only at the evaluation stage rather than silently accepted. Self-supervised and preference-learning approaches to citation generation (e.g., context-attribution alignment techniques and citation-focused preference optimization) attempt to train models to prefer citation patterns that survive automated attribution checks, moving traceability optimization from a purely evaluative to a partly generative concern. Reinforcement-learning approaches explicitly reward-shape generation toward higher citation-recall and citation-precision scores, reporting improved faithfulness relative to supervised-fine-tuning-only baselines, though typically evaluated on single-pass RAG QA rather than multi-stage literature-synthesis pipelines specifically.

4.3 Multi-Agent Architectures for Verification

A distinct and rapidly growing set of current approaches introduces dedicated *verification agents* into the pipeline itself rather than treating traceability evaluation as an external, offline step. In these designs, one agent (or a small agent ensemble) is tasked specifically with checking the citations, extracted claims, or bibliographic metadata produced by upstream generation agents, functioning analogously to the second-reviewer role in traditional systematic-review methodology. Retrieval-augmented validation approaches apply this pattern directly to citation checking, cross-referencing generated citations against retrieval indices before claims are finalized. Semantic-caching and memory-augmented multi-agent designs have additionally been proposed to make such multi-stage, multi-agent verification pipelines computationally and environmentally sustainable at production scale by reducing redundant LLM invocations across repeated verification calls.

4.4 Domain-Specific Automation Tools

In evidence-based medicine specifically, tool ecosystems such as RobotReviewer, ASReview, Covidence, Rayyan, and DistillerSR occupy a complementary niche: rather than generating synthesized prose, they support human reviewers in screening, extraction, and bias assessment, with traceability enforced structurally by requiring every automated suggestion to be paired with the specific source sentence(s) it was derived from, and typically requiring human sign-off before a suggestion is incorporated into the final review (Marshall et al., 2016; Khalil et al., 2022). This human-in-the-loop design pattern, in which the system’s role is confined to prioritization and evidence surfacing rather than autonomous claim generation, represents a conservative point on the traceability-versus-autonomy spectrum that fully agentic synthesis pipelines are

7 Conclusion

Agentic literature-synthesis pipelines offer substantial promise for accelerating evidence-based research, but that promise is contingent on the ability to verify, at scale, that synthesized claims are genuinely traceable to the sources they cite. This paper has argued that evidence traceability is not a single property to be captured by one aggregate score, but a layered set of guarantees—existence, attribution precision/recall, claim-evidence entailment, and cross-agent provenance completeness—each addressing a distinct failure mode documented in the RAG-attribution, citation-hallucination-auditing, and systematic-review-automation literatures we have reviewed. Current tools and benchmarks, including ALCE, RAGAS, ARES, and FActScore, provide well-developed instruments for the first three layers in single-pass or short-form generation settings, while domain-specific audits in medicine and computational linguistics have quantified the scale of the underlying problem. Provenance completeness across multi-stage, multi-agent pipelines, however, remains comparatively underdeveloped, both methodologically and in terms of standardized benchmarks. We hope the taxonomy and synthesis offered here provide a useful reference point for researchers designing evaluation protocols, verification-agent architectures, and reporting standards for the next generation of agentic literature-synthesis systems.

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